Distance Decay as a Primary Transferable Component: Zero-Shot Gravity Model Calibration from Trip-Length Distributions

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Abstract

Predicting commuting flows in cities without access to individual travel trajectories is a central challenge in urban mobility modeling: most high-accuracy methods require local origin-destination (OD) surveys or individual-level records for calibration, limiting their applicability in data-scarce or privacy-regulated environments. This study investigates whether aggregate trip-length distributions—population-level histograms of traveled distances available from mobile network operators and tech platforms—contain sufficient information to recover the city-specific distance-decay structure needed for OD-free gravity model calibration. We distinguish two transfer settings: an *OD-free calibration* setting that uses only target-city aggregate trip-length distributions (but not individual OD records), and a *pure zero-shot fallback* that uses no target mobility data whatsoever. Evaluating across 50 US metropolitan areas (25 source cities for training and 25 held-out target cities for evaluation), we show that decay parameters estimated from aggregate histograms alone correlate strongly with those fitted from complete OD matrices ($\boldsymbol{r = 0.909}$ for the power-law exponent γ), with a downstream prediction gap of only **0.11%** under oracle outflow conditions. Through factorial ablation and Shapley value analysis, our measurements attribute 85.8% of predictive capacity to distance decay within the investigated gravity framework—far exceeding origin production (6.4%) and destination attraction (7.9%)—which we interpret as evidence that distance decay functions as the primary transferable component of spatial interaction within this framework. Under oracle outflow conditions, aggregate-recovered decay achieves downstream performance statistically equivalent to OD-fitted decay. In the fully survey-free setting using independently sourced aggregate data (Meta Movement Distribution Maps), the proposed model achieves CPC 0.708, narrowing the gap to neural zero-shot baselines to 5.8% in physically constrained coastal environments.

Keywords: Urban mobility, distance decay calibration, gravity models, aggregate mobility data, zero-shot transfer, open-data modeling

1 Introduction

Predicting origin-destination (OD) travel flows is essential for urban transport planning and infrastructure design (Barbosa et al. 2018; González et al. 2008; Song et al. 2010). However, traditional travel surveys and proprietary cellular records are highly restricted due to cost, commercial licensing, and privacy concerns. Consequently, a central challenge in urban mobility modeling is

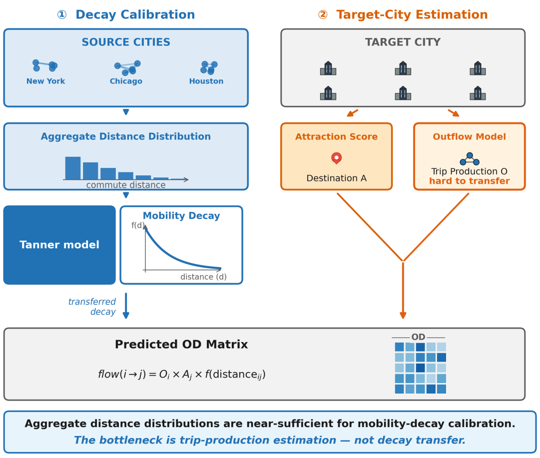

Fig. 1 Overview of the core findings: Aggregate trip-length distributions contain the necessary information to recover city-specific distance decay $f(d)$, which acts as a primary transferable component for predicting zero-shot flows within the investigated gravity framework.

the zero-shot cross-city transfer problem: predicting commuting flows in a target city where no local OD surveys are available.

While recent deep learning approaches achieve high accuracy, they require abundant individual-level OD records for training, limiting their generalizability and raising privacy concerns. In contrast, classical gravity models (Wilson 1971) offer physical interpretability and structural simplicity but require local calibration of their distance-decay functions. This raises a fundamental scientific question: **Can aggregate mobility distributions, which do not reveal individual OD pairs, serve as a reliable calibration signal for recovering city-specific distance-decay functions?**

In this paper, we address this question empirically across 50 US cities under two complementary transfer settings. In the **OD-free calibration setting**, the model uses target-city aggregate trip-length distributions (not individual OD records) to recover distance-decay parameters—this is the primary deployable scenario. In the **pure zero-shot fallback**, the model uses no target mobility data whatsoever, relying only on a national decay prior derived from source cities. We show that **aggregate trip-length histograms contain sufficient information to recover the city-specific distance-decay structure with high fidelity, and our Shapley analysis attributes 85.8% of predictive capacity to distance decay within the investigated framework—consistent with the interpretation that distance decay is a primary transferable component, which helps explain why OD-free calibration achieves competitive performance across the evaluated US metropolitan areas.**

To investigate these relationships systematically, we organize the investigation around three research questions:

RQ1 **(Decay Recovery)** Can aggregate trip-length distributions recover the city-specific distance-decay structure without access to individual OD trajectories?

RQ2 **(Decay Impact)** How does distance decay impact zero-shot gravity transfer of commuting flows, and to what extent does its recovery from trip-length distributions drive transfer performance?

RQ3 **(Morphological Scope)** Under what urban morphological conditions does aggregate-calibrated decay perform most effectively, and what physical geography factors govern those conditions?

Our empirical investigation across 50 US metropolitan areas provides three specific contributions:

1. **Decay Recovery from Aggregate TLD (RQ1)**: We demonstrate that aggregate trip-length distributions can successfully recover the dominant shape of city-specific distance decay, especially the power-law exponent γ ($r = 0.909$), achieving a near-identical downstream CPC (0.11% average gap) when target production totals are controlled using oracle outflows.
2. **Component-Level Shapley Analysis (RQ2)**: Our game-theoretic Shapley analysis shows that distance decay dominates predictive capacity gain (accounting for 85.8% of the total gain) specifically within this decomposed gravity game, rather than representing a universal claim across all urban mobility model structures.
3. **Exploratory Morphology Analysis (RQ3)**: We provide exploratory analysis suggesting that physics-constrained gravity models perform relatively better in physically constrained coastal cities (narrowing the gap

to DeepGravity to 5.8%), though we treat these findings as preliminary due to the small morphological group sizes.

These findings indicate that trip-length distributions encode the distance-decay structure needed for OD-free calibration, and that identifying and recovering this primary transferable component is sufficient to recover the majority of the transferable signal within the evaluated gravity framework. The resulting model achieves competitive performance across most urban morphologies, though a performance gap versus high-capacity neural baselines remains (5.8%–10.3% depending on city type). A pure zero-shot fallback using only a national decay prior (with no target mobility data) retains 92.4% of oracle performance, providing an additional deployment pathway for severely data-scarce settings. The rest of the paper provides systematic empirical verification organized around these three research questions.

2 Related Work

2.1 Classical and Structured Mobility Models

Gravity models (Wilson 1971; Tanner 1961) and radiation models (Simini et al. 2012; Alis et al. 2021) provide the basis of spatial interaction modeling. Gravity models employ physical metaphors to link flows to mass and distance in zones, while radiation models (Stouffer 1940) interpret the choice of destinations probabilistically on the basis of intervening opportunities. Scaling-law studies (Brockmann et al. 2006; González et al. 2008) and predictability analysis (Song et al. 2010) have shown that travel distances obey clear statistical patterns in urban settings. Both model classes offer interpretability and parsimony, but gravity models require local OD data to calibrate decay parameters, while radiation models underperform in cities where the intervening opportunities assumption breaks down due to strong directional commuting preferences or constrained network topologies.

2.2 Deep Learning for OD Prediction

DeepGravity (Simini et al. 2021) uses multilayer perceptrons to associate urban characteristics with flow volumes. Recent advancements have included graph neural networks (Rong et al. 2023), spatio-temporal networks (Wang et al. 2021), and OSM-based commuter modeling (Atwal et al. 2025). These methods achieve strong predictive accuracy where training data is abundant, but require substantial target-city OD records for training and fine-tuning, limiting their applicability in data-scarce environments and raising concerns about out-of-distribution generalizability and model interpretability.

2.3 Aggregate Mobility Data for Decay Calibration

Today's location-aware mobile phones yield mobility signals of entire populations (Barbosa et al. 2018; González et al. 2008) that can be summarized to form trip-length distributions—histograms of traveled distances—that do not reveal any OD pairs (Pappalardo et al. 2023). Previous studies have proven that such distributions are indicative of the friction of urban trips and match distance-decay models (Brockmann et al. 2006; Lenormand et al. 2012, 2016). However, to the best of our knowledge, there is no existing study on using such distributions to calibrate zero-shot cross-city transfer.

2.4 Cross-City Transfer Learning

Transfer learning across cities is designed to forecast mobility in cities that lack OD survey data. The existing literature on cross-city transfer learning can be divided into three categories:

1. **Similarity-based**: RegionTrans (Wang et al. 2018) uses spatial similarity to align source and target regions; CrossTReS (Jin et al. 2022) reweights source regions using meta-learning.
2. **Representation learning**: Geospatial embeddings and GeoAI foundation models (Mai et al. 2022; Mendieta et al. 2023) use representation learning to develop features for urban systems.
3. **Domain adaptation**: STAN (Wang et al. 2022) and meta-spatiotemporal networks (Yao

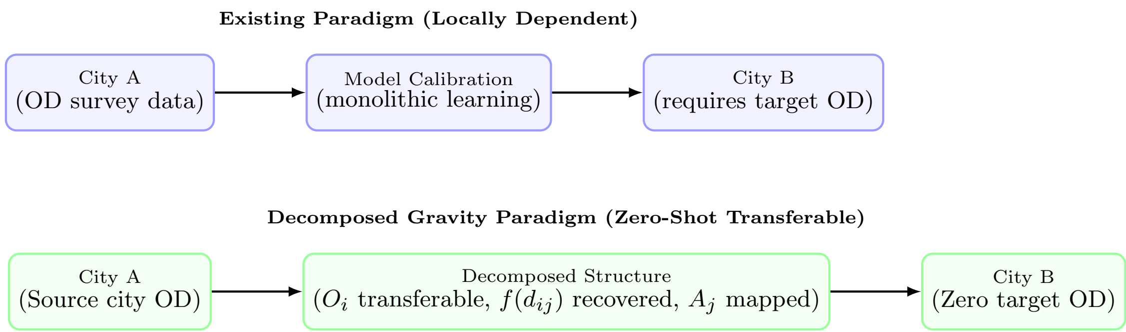

Fig. 2 Conceptual comparison of the existing locally dependent modeling paradigm versus the decomposed gravity paradigm investigated in this study. Under the existing paradigm, models require target-city OD observations to retrain or calibrate parameters. In the decomposed paradigm, transferable origin production structures are learned from source cities, decay is recovered from target aggregate data, and attraction is mapped from local open features, enabling zero-shot transfer without target-city OD surveys.

et al. 2019) align trained models to the target domains.

All three categories require access to target-city observations, fine-tuning, or domain-specific calibration.

2.5 Quantifying the Relative Role of Distance Decay in Transfer

Although distance decay is widely acknowledged as central to gravity models, its relative importance compared to other components (origin production, destination attraction) in the context of zero-shot cross-city transfer has not been systematically quantified. SHAP (Lundberg and Lee 2017) offers a theoretically grounded framework for decomposing predictive contributions, but its application to cross-city component analysis in mobility models remains unexplored.

2.6 Research Gap and Positioning

Table 1 compares how existing model classes address the core challenge of this study: calibrating distance decay from aggregate summaries that reduce privacy risks compared with individual trajectories for zero-shot cross-city transfer.

The table reveals three specific gaps that this study addresses, each directly corresponding to one of the three research questions:

G1 (RQ1): **No aggregate-based decay calibration for zero-shot transfer.** Classical gravity and transfer methods require target-city OD observations for calibration. No existing work demonstrates that aggregate trip-length summaries, which reduce privacy exposure compared with individual trajectories, can recover target decay parameters for zero-shot transfer.

G2 (RQ2): **The impact of distance decay on zero-shot transfer has not been quantified.** The relative contribution of distance decay to zero-shot transfer performance (compared to production and attraction) remains unquantified, leaving its precise role relative to other components unclear.

G3 (RQ3): **Urban morphological conditions for aggregate calibration are unknown.** The impact of urban morphology and physical geography on aggregate calibration accuracy and the role of distance decay has not been systematically investigated.

3 Problem Setting

To establish a rigorous evaluation framework, we define three distinct settings of target-city data availability, representing different operational constraints and analytical diagnostics:

1. **Setting 1: Pure Zero-Shot Transfer**: In this setting, absolutely no target-city mobility or travel survey data (neither OD matrix nor aggregate TLD) is available. The model must rely entirely on open spatial data (OSM, WorldPop) for target features, and utilizes a fixed *National Prior* for the distance-decay function derived from training cities. This represents the fallback scenario under extreme data scarcity.

Table 1 Comparative analysis of human mobility models. "Requires Target OD" indicates whether the model requires zone-to-zone target-city commuting matrices for calibration or training. "Uses Aggregate TLD" specifies if the model leverages aggregate 1D trip-length distributions. "Cross-City Zero-Shot" indicates if the model can be deployed to unseen cities without target-city OD data.

Model Class	**Requires Target OD?**	**Uses Aggregate TLD?**	**Interpretable Decay?**	**Cross-C**
Gravity (Wilson 1971)	✓	×	✓	
Radiation (Simini et al. 2012)	×	×	×	
DeepGravity (Simini et al. 2021)	×*	×	×	
GNN OD (Rong et al. 2023)	✓	×	×	
RegionTrans (Wang et al. 2018)	✓	×	×	
This study (Proposed OD-Free)	×	✓	✓	

* DeepGravity requires source-city OD records for training but not for target-city inference, analogous to the GBDT outflow model in this study.

2. **Setting 2: OD-Free Aggregate Calibration**: This is the primary proposed operational pipeline. We assume access to target-city aggregate trip-length distributions (TLD) under the form of 1D distance-bin histograms (e.g., Meta Movement Distribution Maps). Individual zone-to-zone commuting trajectories (OD matrices) remain completely unavailable. This setting enables city-specific decay calibration while maintaining high privacy protection and zero survey requirements.
3. **Setting 3: Oracle Diagnostic Baseline**: This is a controlled baseline used exclusively for component-level analysis and algorithm validation (Experiment A). The target city's true origin outflows (O_i) are used as oracle values to isolate the distance-decay estimation process from potential prediction errors in the trip production component.

4 Methodological Design

The approach taken in this study does not attempt to learn the complete urban mobility process end-to-end. Instead, the central task is to identify the transferable component of spatial interaction within the investigated decomposed gravity framework—distance decay—and recover it from aggregate trip-length distributions. Origin production and destination attraction are handled via cross-city transfer learning and an open-data heuristic respectively, but the recovery of the distance-decay structure from aggregate data is the core contribution that makes zero-shot calibration possible.

4.1 Decomposition Hypothesis

We hypothesize that OD flows can be decomposed into three separable, city-independent components:

$$T_{ij} = O_i \cdot \frac{A_j f(d_{ij})}{\sum_k A_k f(d_{ik})} \tag{1}$$

where O_i represents origin production capacity, $f(d_{ij})$ represents distance decay, and A_j captures destination attraction. We hypothesize that: (1) each component is approximately independent, (2) the underlying mechanisms can be learned from spatial descriptors of the built environment, and (3) the learned origin production relationship transfers directly to unseen target cities, while decay parameters are recovered from target-city aggregate trip-length distributions, and attraction is computed from locally available open data. This decomposition hypothesis is directly tested by evaluating which of these three components exerts the dominant influence on predictive accuracy. Section 5 evaluates and validates this hypothesis empirically.

4.2 Aggregate-Based Decay Calibration

The central methodological insight of this study is that the aggregate trip-length distribution (TLD) $p(d)$ — a histogram of how many trips occur at each distance bin, summed across all origin-destination pairs in a city — represents the marginal distribution of the OD matrix over distance (the decay function f_θ used here is the

Tanner form defined in Section 3.3). Formally:

$$p(d) = \sum_{i,j} T_{ij} \cdot \mathbf{1}[d_{ij} \approx d] \tag{2}$$

In principle, the marginal $p(d)$ does not automatically represent a sufficient statistic for the decay parameters, as the number of trips observed at distance d is strongly conditioned by a city's spatial opportunity distribution (the spatial arrangement of population O_i and jobs A_j), zone geometries, and boundary effects. To address this and prevent bias from spatial configurations, we formalize the recovery of decay parameters $\theta = (\gamma, \beta)$ using an exposure-corrected likelihood model.

Let y_b be the observed commuter count in distance bin $b \in \{1, \ldots, K\}$, obtained from the target-city aggregate trip-length histogram. Under the production-constrained gravity framework, the predicted flow between zones i and j under decay parameters θ is formulated as:

$$\hat{T}_{ij}(\theta) = O_i \frac{A_j f_\theta(d_{ij})}{\sum_k A_k f_\theta(d_{ik})} \tag{3}$$

where O_i is the target outflow (oracle in Experiment A; GBDT-estimated in Experiments B and C), A_j is the open-data attraction heuristic (Section 3.4), and f_θ is the Tanner decay function (Section 3.3). The predicted probability of a trip falling into distance bin b is the sum of predicted flows in that bin normalized by total predicted trips:

$$P_\theta(b) = \frac{\sum_{i,j: d_{ij} \in b} \hat{T}_{ij}(\theta)}{\sum_{i,j} \hat{T}_{ij}(\theta)} \tag{4}$$

This predicted probability $P_\theta(b)$ explicitly incorporates zone geometry, spatial opportunity distribution (A_j), predicted trip generation (O_i), and the constraint structure, directly correcting for distance exposure. Note that the adaptive anchor parameter δ is fixed statically based on zone geometries (Equation 7) prior to calibration; thus, the MLE optimization search space is restricted exclusively to the decay parameters $\theta = (\gamma, \beta)$. Assuming a multinomial distribution of trips across the distance bins, we estimate the decay parameters $\theta = (\gamma, \beta)$ by maximizing the multinomial log-likelihood (or equivalently, minimizing the Kullback-Leibler divergence) between the observed histogram fractions $\bar{y}_b = y_b / \sum_{b'} y_{b'}$ and the predicted probabilities $P_\theta(b)$:

$$\mathcal{L}(\theta) = \sum_{b=1}^{K} \bar{y}_b \log P_\theta(b) \tag{5}$$

Minimizing $-\mathcal{L}(\theta)$ recovers the parameters θ from the target-city aggregate TLD without requiring individual zone-to-zone OD flow records, explaining why aggregate, non-OD-revealing histograms are sufficient for high-fidelity decay calibration in practice. The decomposed gravity formulation used in this study (referred to as the separable decomposed formulation for zero-shot transferable flows) is illustrated in Fig. 3.

4.3 Distance Decay Function

We use the Tanner multiplicative deterrence function (Tanner 1961), which combines power-law and exponential terms:

$$f(d_{ij}; \beta, \gamma, \delta) = (d_{ij} + \delta)^{-\gamma} \cdot \exp(-\beta\, d_{ij}) \tag{6}$$

where $\beta > 0$ is the exponential decay rate governing long-range trip attenuation, $\gamma \geq 0$ is the power-law exponent modeling short-to-medium range trip distribution, and $\delta > 0$ is the adaptive geometric anchor.

The offset δ prevents singularity at $d_{ij} = 0$ and adapts to zone size:

$$\delta = 0.5 \times \bar{R}_{\text{zone}}, \qquad \bar{R}_{\text{zone}} = \frac{1}{N} \sum_{i=1}^{N} \sqrt{\frac{\text{Area}_i}{\pi}} \tag{7}$$

where $\bar{R}_{\text{zone}}$ is the mean zone radius. Parameters (β, γ) are estimated via MLE on the aggregate distance-bin distribution. Calibration and optimization details are in the Appendix. Because only binned trip-length histograms are used—not individual OD records—this procedure is compatible with strict data-governance regulations (e.g., GDPR). For numerical implementation and computational efficiency, the theoretical offset δ is regularized by clamping the minimum distance to a small epsilon ($d_{\min} = 10^{-6}$), which yields mathematically equivalent results as $d_{ij} \gg d_{\min}$ for all cross-zone flows ($i \neq j$).

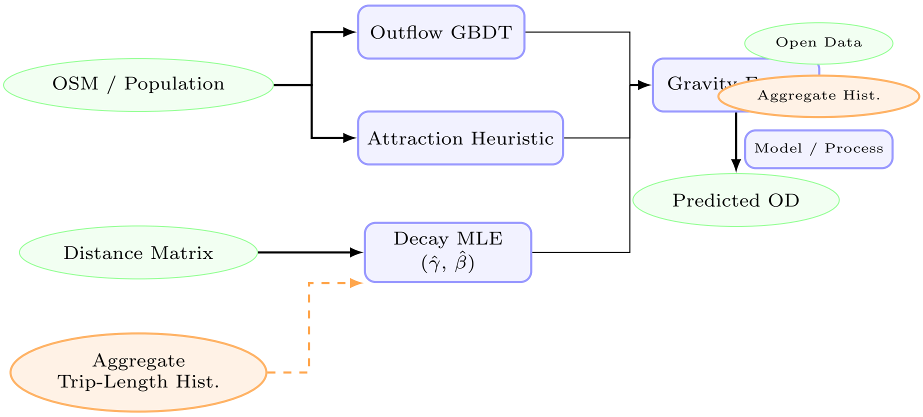

Fig. 3 Overview of the aggregate-calibrated gravity model structure. Globally available open-data features (green ellipses) are used to predict trip production (O_i) via a GBDT model and to compute destination attraction (A_j) via a training-free min-max heuristic. The **Aggregate Trip-Length Histogram** (orange ellipse, dashed arrow)—which provides aggregate summaries that significantly reduce privacy exposure compared with individual trajectories—feeds the Decay MLE block to recover city-specific parameters ($\hat{\gamma}, \hat{\beta}$) without requiring any individual OD trajectories. The three components are combined via multiplicative gravity fusion to predict zero-shot flow matrices (T_{ij}).

4.4 Origin Outflow Component (O_i)

The trip production $O_i = \sum_j T_{ij}$ is the total trips originating from zone i. We predict O_i with a GBDT regressor trained on a parsimonious set of **6 core macroscopic features**: total population (P_i), total POI count (POI_i), zone area (Area_i), population density ($\rho_{pop,i}$), POI density ($\rho_{poi,i}$), and road density (rd_i). This parsimonious feature set is chosen based on dimensional scaling in urban physics and Occam's razor (see Section 4), and avoids overfitting to source-domain geometries. Hyperparameters are in Appendix C.

4.5 Destination Attraction Component (A_j)

The destination attraction component, A_j, represents the capacity of zone j to attract trips, acting as a proxy for local spatial opportunities (e.g., employment, commerce, and residential destinations) (Wilson 1971; Simini et al. 2012). In traditional gravity models, A_j is typically estimated using supervised regression models or calibrated zone-specific parameters. However, in zero-shot cross-city transfer scenarios where target-city commuting flows are completely unavailable, supervised estimation of destination attraction poses a significant risk of overfitting to the source domain.

To address this challenge and ensure robust zero-shot generalizability, we formulate a training-free, scale-invariant opportunity heuristic $\hat{A}_j$. This heuristic combines two distinct dimensions of spatial opportunities: residential capacity (reflecting return-to-home flows) and activity-based opportunities (reflecting work, shopping, and leisure destinations). Formally, we define:

$$\hat{A}_j = \frac{1}{2}\left[\mathrm{minmax}(P_j) + \mathrm{minmax}(\mathrm{POI}_j)\right] \tag{8}$$

where P_j represents the gridded population count of zone j (obtained from open demography), and POI_j represents the total count of commercial, industrial, and social Points of Interest (POIs) extracted from OpenStreetMap.

Because population and POI counts operate on vastly different numerical scales across heterogeneous metropolitan areas, we apply a min-max scaling function to project both features into a standardized interval $[0, 1]$:

$$\mathrm{minmax}(x_j) = \frac{x_j - \min_{k\in\mathcal{V}} x_k}{\max_{k\in\mathcal{V}} x_k - \min_{k\in\mathcal{V}} x_k + \epsilon} \tag{9}$$

where $\mathcal{V}$ denotes the set of all zones within the target metropolitan area, and $\epsilon = 10^{-9}$ is a small positive constant to prevent division by zero. This scaling preserves the relative hierarchy of attractiveness within a city while normalizing absolute scale differences between different metropolitan areas. In Section 5, the ablation analysis (Table 5) validates that this training-free heuristic contributes meaningfully to zero-shot performance, with its removal causing a 4.4% CPC drop, confirming that the simplified formulation is sufficient without introducing source-domain overfitting.

4.6 Algorithmic Flow

The complete pipeline for training and zero-shot deployment is formalized in Algorithm 1.

4.7 Data Firewall and Zero-Leakage Protocol

To guarantee that target-city commuting matrices are never leaked into the model components, we implement a strict data firewall between source-city and target-city variables across all phases of the transfer learning pipeline:

1. **Outflow Predictor Firewall**: The GBDT outflow predictor is trained exclusively on built-environment features and ground-truth trip counts from source cities. Target-city spatial features are only used during inference, ensuring that the outflow predictions contain no target OD flow information.
2. **Decay Parameter Firewall**: City-specific distance-decay parameters $(\hat{\gamma}, \hat{\beta})$ are recovered solely from the target city's aggregate 1D trip-length distribution (Setting 2). No source-city parameters or target-city zone-to-zone flows are mixed into this process.
3. **Attraction Heuristic Firewall**: The attraction heuristic uses only the target city's population and POI feature hierarchies via a training-free min-max function, transferring no parameters from source cities.
4. **Evaluation Firewall**: Target-city zone-to-zone OD matrices are kept completely blind to all model components, used solely as the test ground-truth to calculate CPC and RMSE.

5 Data and Transferable Urban Representation

5.1 Design Principle: Open-Data-Compatible

All predictive features in the proposed approach are drawn from globally available open data rather than proprietary cellular records or household surveys, making the methodology deployable in any city. We intentionally exclude proprietary localized employment registries, which lack global equivalents, and instead proxy employment activity through commercial and industrial POI densities from OpenStreetMap. Population counts in this study are sourced from the third-party mobility dataset; an open-data alternative using WorldPop (Tatem 2017) gridded estimates is validated in Appendix A.2 and yields comparable results, confirming that the methodology can be operated in a fully open-data mode.

5.2 Data Sources

We use three data sources. Their strict isolation across the train, calibrate, predict, and evaluate phases is summarized in Table 2.

Ground-truth origin-destination (OD) flows between census tracts are obtained from a third-party human mobility dataset covering 50 US metropolitan areas. For the distance-decay prior, we evaluate two aggregate sources: (1) a **third-party aggregate baseline**, in which the same third-party OD dataset is aggregated into distance-bin histograms—this serves as an oracle bound, not a claim of data independence; and (2) the **Meta Movement Distribution Maps** (Meta 2026), independently collected county-level aggregate statistics that do not derive from the same ground-truth OD source—this is the true OD-free, independently sourced calibration scenario. In both cases, the aggregate prior does not provide zone-to-zone OD matrices. The OD matrix serves as the ground-truth target for city-specific evaluation and as training labels for source cities' production models; it is never used as a predictive feature for target prediction. By showing that independently sourced aggregate decay curves (Meta MDM) yield downstream performance comparable to full OD matrices (Section 5), we provide empirical evidence that publicly released aggregate mobility products may serve as a viable alternative for calibrating spatial interaction models, avoiding the need for proprietary or privacy-invasive individual surveys in practical planning pipelines.

Algorithm 1: Aggregate Calibration and Zero-Shot Prediction

Input: Source cities data $\{\mathbf{X}^{(c)}, \mathbf{T}^{(c)}\}_{c=1}^{C}$; Target city spatial features $\mathbf{X}^{(t)}$; Target city aggregate trip-length histogram $\{y_b\}_{b=1}^{K}$ (or national prior if unavailable)
Output: Predicted Target City OD Matrix $\hat{\mathbf{T}}^{(t)}$

Phase 1: Offline Training (Source Cities)
1. **Production Model Training**: Train the outflow model $g(\mathbf{x})$ (GBDT) on pooled source-city spatial features $\mathbf{X}^{(c)}$ and ground-truth outflows $O_i^{(c)}$.

Phase 2: Online Calibration & Zero-Shot Transfer (Target City)
2. **Decay Calibration**: Recover target-city decay parameters $(\hat{\beta}, \hat{\gamma})$ via MLE on the target city's aggregate distance-bin histogram $\{y_b\}_{b=1}^{K}$. (If unavailable, fall back to the source-derived national prior $(\beta_{\text{nat}}, \gamma_{\text{nat}})$).
3. **Attraction Mapping**: Compute target city zone attractions using the training-free heuristic: $\hat{A}_j^{(t)} = \frac{1}{2}\left(\text{minmax}(P_j^{(t)}) + \text{minmax}(\text{POI}_j^{(t)})\right)$.
4. **Zero-Shot Target Prediction**:
   a. Predict target city outflows: $\hat{O}_i^{(t)} = g(\mathbf{x}_i^{(t)})$.
   b. Compute adaptive target offset based on zone sizes: $\delta^{(t)} = 0.5 \times \bar{R}_{\text{zone}}^{(t)}$.
   c. Fuse components and normalize outflows: $\hat{T}_{ij}^{(t)} = \hat{O}_i^{(t)} \frac{\hat{A}_j^{(t)}(d_{ij}+\delta^{(t)})^{-\hat{\gamma}} e^{-\hat{\beta} d_{ij}}}{\sum_k \hat{A}_k^{(t)}(d_{ik}+\delta^{(t)})^{-\hat{\gamma}} e^{-\hat{\beta} d_{ik}}}$.

Table 2 Data leakage protocol mapping the strict isolation between source-city and target-city datasets across training, calibration, prediction, and evaluation phases.

Phase	Source Cities Data	Target Cities Data	Leakage Firewall
Outflow Model Training ($g(\mathbf{x})$)	Built-environment features & Ground-truth OD outflows ($O_i^{(c)}$)	None	Strict block: Target features/flows are never seen.
Decay Calibration ($\hat{\gamma}, \hat{\beta}$)	None	Aggregate Trip-Length Distribution (TLD)	Strict block: Target zone-to-zone OD flows are never accessed.
Attraction Mapping (A_j)	None	Built-environment features (OSM, WorldPop)	Training-free heuristic; no parameter transfer.
Downstream Prediction ($\hat{T}_{ij}$)	Trained Outflow GBDT model parameters	Target spatial features & Calibrated decay parameters	Zero-shot transfer; target OD matrix remains completely blind.
Evaluation	None	Ground-truth OD flows (T_{ij})	Used solely for testing; never fed back to model.

5.2.1 OSM-Derived Features

We extract POIs and road features using the Overpass API from OpenStreetMap snapshots (Atwal et al. 2025). POI counts are calculated across 16 land-use categories (e.g., retail, industrial, education, health).

5.2.2 Population Data

Population counts are sourced from the third-party mobility dataset used for ground-truth OD flows. As validated in Appendix A.2, they can be replaced with WorldPop (Tatem 2017) gridded estimates (100 m resolution), which is available globally, yielding comparable results.

5.3 Transferable Urban Features

We use a two-stage procedure—feature engineering followed by stability selection—to identify spatial patterns that generalize across cities. The feature taxonomy is shown in Fig. 4.

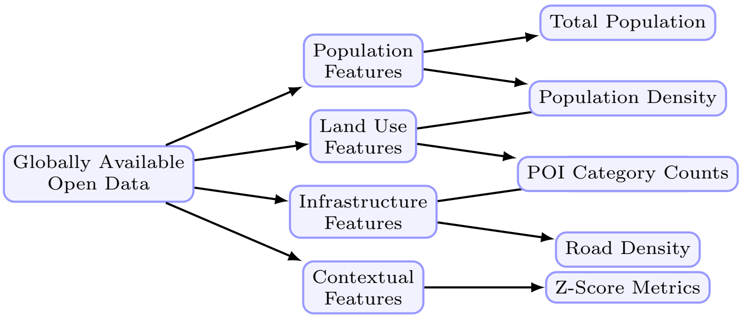

Fig. 4 Taxonomy of the globally available open data features used in the outflow model. Features are categorized into four dimensions: population, land use, infrastructure, and contextual scale factors.

5.3.1 Candidate Feature Set (30 dimensions)

We initially construct 30 candidate features per zone. This candidate space comprises 23 scale and density variables (zone area, road length, residential population, total POI counts, and 16 POI categories) and 7 city-wide Z-score context features. These 7 context features are constructed by applying Z-score normalization to the 7 key population and infrastructural density features that exhibit high variance across cities. The Z-score context features are defined as:

$$z_{ip} = \frac{x_{ip} - \mu_p^{(c)}}{\sigma_p^{(c)}} \tag{10}$$

where $\mu_p^{(c)}$ and $\sigma_p^{(c)}$ represent the mean and standard deviation of feature p across all zones in city c. These context features normalize absolute scale differences, capturing the relative hierarchy of zones within their respective metropolitan areas.

5.3.2 SHAP-Based Feature Stability Selection

To identify features that generalize across cities, we apply SHAP importance analysis (Lundberg and Lee 2017) to the GBDT production model. Features are ranked by a cross-city stability score $S_p = \mu_p / (\sigma_p + \epsilon)$, where μ_p and σ_p denote the mean and standard deviation of per-city GBDT SHAP importances and $\epsilon = 10^{-5}$. A feature is retained if $\mu_p \geq 0.005$ and $S_p \geq 1.0$, ensuring both relevance and cross-city consistency. This analysis identifies 23 stable features from the 30-candidate space (23.3% dimensionality reduction) and confirms that the 6 core structural features used in the deployed model (Section 3.4) are among the primary and most stable predictors, with no accuracy loss relative to the full candidate set (Appendix C). *Note: this feature-level analysis is distinct from the inter-component Shapley game in Section 5.2, which quantifies the relative contribution of the three model components $f(d)$, O_i, and A_j.*

5.4 Urban Morphology Grouping (RQ3)

To investigate RQ3—under what morphological conditions aggregate-calibrated decay is most effective—the 25 held-out target cities are classified into four morphological groups based on physical geography and urban structure. The grouping follows established urban morphology criteria (González et al. 2008; Brockmann et al. 2006):

Dense / Transit (5 cities): High-density urban cores with extensive rapid transit networks (e.g., subway, light rail), where multi-modal options decouple travel distance from actual trip cost. Distance friction is weakest in these environments. *Cities: Boston, Long Beach, New York, Philadelphia, Washington DC.*
Sprawling / Grid (10 cities): Low-to-medium density cities built on regular road grids, with car-dependent mobility patterns. Distance decay is moderate and relatively uniform. *Cities: Atlanta, Dallas, El Paso, Houston, Jacksonville, Las Vegas, Mesa, Omaha, San Antonio, Tulsa.*
Polycentric (7 cities): Cities with multiple distinct employment and activity centers, where flows cluster around nodes rather than following simple distance gradients. *Cities: Albuquerque,*

Charlotte, Detroit, Louisville, Milwaukee, Raleigh, San Jose.
Coastal (3 cities): Cities where coastlines or topographic barriers physically constrain the road network, creating strong geographic separation between zones. The constrained geography is hypothesized to align with a stronger influence of physical distance friction, making structured models highly effective. *Cities: Portland, Seattle, Virginia Beach.*

This grouping enables direct investigation of how physical geography modulates the role of distance decay as a transferable component, providing empirical scope for the aggregate calibration approach.

5.5 Cross-City Transfer Evaluation Protocol

Standard mobility evaluations split OD pairs within a single city, which leaks spatial information from test zones into training. We instead use a strict **25/25 Stratified Split**: The 50 US metropolitan areas are divided into two disjoint sets of 25 training source cities and 25 held-out target cities. Models are trained exclusively on pooled data from the source cities and evaluated on the held-out target cities without any access to target-city OD data or retraining. The split is stratified by geographic region and urban morphology to ensure statistical representation across city types.

5.5.1 Information Leakage Prevention

Under this cross-city evaluation protocol, we enforce a strict information firewall between the source and target environments.

- **Forbidden**: Target-city flow matrices T_{ij}, target-city household survey results, target cellular data, and target trip-demand signals.
- **Allowed**: Centroid coordinates, administrative zone boundaries, OpenStreetMap POIs, and gridded open demography (WorldPop).

Since destination attraction is computed via a training-free open-data heuristic and distance decay parameters are recovered from target-city aggregate trip-length distributions (or fall back to source-trained national priors), zero-shot leakage of target-city OD flows is mathematically prevented.

5.5.2 Evaluation Metric: Common Part of Commuters

We measure prediction quality with the Common Part of Commuters (CPC) (Lenormand et al. 2012):

$$\mathrm{CPC}(T, \hat{T}) = \frac{2\sum_{i,j}\min(T_{ij}, \hat{T}_{ij})}{\sum_{i,j} T_{ij} + \sum_{i,j} \hat{T}_{ij}} \tag{11}$$

where T_{ij} represents the ground-truth commuter flow from zone i to zone j, and $\hat{T}_{ij}$ is the predicted flow. The CPC index ranges from 0 (indicating completely disjoint flow matrices) to 1 (indicating perfect prediction).

5.6 Preprocessing & Normalization

Data preprocessing details are summarized in Table 8 in Appendix A.3.

6 Results

6.1 Validation Protocol Overview

To systematically evaluate the proposed approach, we organize our experiments into three distinct phases:

Experiment A: Decay Parameter Recovery Only (RQ1)
We isolate the decay component by fixing origin production O_i to oracle outflows and attraction A_j to the open-data heuristic. We then evaluate whether aggregate trip-length distributions (TLD) can recover decay parameters (γ, β) with high fidelity compared to parameters fitted directly on the ground-truth OD matrices.
Experiment B: Downstream Survey-Free Flow Prediction (RQ2)
We evaluate the proposed decomposed gravity model in a fully deployable, survey-free transfer setting. This model uses GBDT-predicted outflows O_i, the training-free attraction heuristic A_j, and aggregate-recovered (Meta MDM) decay parameters. We compare its prediction performance against zero-shot deep learning baselines (DeepGravity) and locally-calibrated gravity models across 25 held-out target cities.

Experiment C: Exploratory Urban Morphology Analysis (RQ3) We evaluate the performance of our physical gravity formulation across different urban morphological configurations (coastal, sprawl, dense transit, polycentric). Given the small sample size within some categories (e.g., 3 coastal cities), these findings are treated as exploratory.

All models are trained on 25 source cities and evaluated on 25 held-out targets, with no target-city OD data used in the transfer pipelines.

6.2 Experiment A: Decay Parameter Recovery (Answering RQ1)

We begin by evaluating whether trip-length distributions alone contain sufficient information to recover the distance-decay structure (Experiment A, addressing RQ1). This question is foundational: if the decay structure is recoverable from aggregate data, and if distance decay proves to be a primary transferable component (Section 5.2), then aggregate calibration becomes a viable path to zero-shot prediction.

6.2.1 Decay Recovery Results

To isolate the effect of the decay calibration strategy, we fix O_i to oracle outflows and A_j to the OSM heuristic, then compare two modes of estimating (γ, β):

- **Ground Truth (GT)**: Parameters (γ, β) are fitted directly on individual origin-destination (OD) pairs.
- **Recovered**: Parameters are fitted on aggregated distance-bin distributions (trip-length marginal histogram, 20 equal-width bins) without access to individual OD pairs.

Crucially, evaluating downstream OD prediction under oracle outflow (isolating the impact of the decay function) demonstrates that using the recovered parameters yields performance close to the ground truth parameters (Table 3). The mean CPC is 0.7411 under ground truth parameters and 0.7403 under recovered parameters, resulting in an extremely small average gap of just 0.0008 (0.11%). This suggests that aggregate-based recovery can successfully support calibration in downstream flow prediction without requiring individual OD trajectories. Note that this comparison is conducted under oracle outflow (isolating the decay component; the full survey-free model using the national decay prior (CPC 0.6896, see ablation Table 5) uses GBDT-estimated production; the deployed variant using independently sourced Meta MDM decay reaches CPC 0.6860 (Table 6), indicating that production estimation remains the binding constraint in a fully survey-free deployment.

The parameter recovery is strong for γ ($r = 0.909$, $r^2 = 0.826$) but only moderate for β ($r = 0.427$, $r^2 = 0.182$). The weaker recovery of β likely reflects the sensitivity of the exponential term's tail behavior to coarse histogram binning, which compresses information at longer travel distances. Critically, the negligible downstream CPC gap of 0.11% implies that β recovery errors do not substantially distort the decay curve within the commuting distances observed in these US cities—not that β is generally unimportant, but that its effect is not differentiated over the typical inter-zone distance range evaluated here.

6.2.2 Validation via Real-World Meta Mobility Prior (RQ1)

To verify the practical feasibility of using tech-platform telemetry for survey-free modeling, we evaluate the decay parameters recovered from Meta's actual county-level Movement Distribution Maps prior (`Meta MDM Prior (3-Bin)`) on the 25 held-out target cities. We compare its downstream CPC performance under oracle outflows against models calibrated on third-party mobility databases aggregated into 3 bins (`Third-Party Baseline (3-Bin)`) and 20 equal-width bins (`Third-Party Oracle (20-Bin)`), as well as the supervised zero-shot DeepGravity baseline (shown as a reference point; DeepGravity is an end-to-end model that does not use oracle outflows).

As shown in Table 4, the survey-free `Meta MDM Prior (3-Bin)` calibration yields a mean CPC of 0.7080 across the 25 target cities. This represents a minor performance gap of only −4.36% compared to the `Third-Party Oracle (20-Bin)` survey oracle (0.7403), which requires complete target-city commuting matrices. When compared to the same 3-bin resolution baseline (`Third-Party Baseline (3-Bin)`, 0.7348), the gap narrows to −3.65%. This minor performance trade-off suggests that publicly released aggregate location fractions, such as those collected by

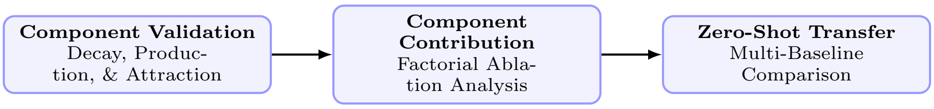

Fig. 5 Evidence supporting RQ1–RQ3 showing the three evaluation stages used to systematically verify the aggregate calibration approach.

Table 3 Decay parameter recovery performance metrics and downstream CPC under oracle outflow across the 25 target cities (RQ1).

Parameter / Strategy	MAE	RMSE	Pearson r	Pearson r^2	Downstream CPC
Power-law exponent (γ)	0.0347	0.0463	0.909	0.826	–
Exponential decay rate (β, km^{-1})	0.0026	0.0058	0.427	0.182	–
Ground Truth (GT) Decay	–	–	Reference	–	0.7411
Recovered (Proposed) Decay	–	–	(see r above)	–	0.7403 (Gap: -0.11%)

Meta, may serve as a viable calibration signal in place of traditional household surveys. The small difference between the Meta MDM 3-bin performance (0.7080) and the Third-Party 3-bin baseline (0.7348) stems from differences in bin boundary definitions; the Meta MDM uses unequal physical thresholds designed for privacy protection, whereas the Third-Party baseline is binned using idealized equal-width intervals.

Transparency note on CPC values: The CPC values reported in Table 4 (e.g., DeepGravity at 0.7727 and the Oracle baseline at 0.7403) are computed specifically over the subset of target cities where Meta MDM data is available. These values differ from the results in our primary baseline comparison in Section 5.3 (Table 6), which are evaluated over the complete 25 held-out target cities in our 25/25 split. In the full split, DeepGravity E2E zero-shot reaches 0.7529 and the proposed Survey-Free model (with Meta MDM decay) reaches 0.6860.

To evaluate the model's robustness under extreme data sparsity where even aggregate target-city distributions are missing, we evaluate the National Decay Fallback strategy (using parameters $\gamma = 1.5, \beta = 0.05$ derived from source cities). This fallback yields a mean CPC of 0.6839 across the 25 target cities, retaining 92.4% of the Third-Party Oracle (20-bin) performance (0.7403) and strongly outperforming the Uniform baseline (0.5290) which completely omits distance decay. This confirms that even a coarse national decay fallback offers high predictive utility under severe data sparsity.

6.2.3 Decay Bin Sensitivity Analysis (RQ1)

Sweeping the number of distance bins $K \in \{3, 5, 10, 20\}$ demonstrates that downstream CPC stabilizes rapidly: mean CPC is 0.7346 at $K = 3$, 0.7367 at $K = 5$, 0.7391 at $K = 10$, and 0.7398 at $K = 20$, compared to the full OD ground truth of 0.7411. This indicates that even extremely coarse histograms (K = 5–10 equal-width bins) are sufficient to robustly capture the city-specific decay profile. Full sensitivity details and the recovery error heatmap are in Appendix A.2.

6.2.4 Origin Production Transferability

This subsection completes the component-level validation of O_i alongside the decay and attraction validations above. Results feed directly into the integrated multi-baseline comparison in Experiment B (Section 5.3).

To isolate the contribution of O_i, we hold A_j (OSM heuristic) and (γ, β) (national prior) fixed, then compare three outflow strategies across 25 held-out cities: a naive population baseline ($O_i \propto \text{Population}_i$), GBDT-predicted outflows, and oracle outflows.

We find that GBDT outflows (CPC 0.6893) beat the naive population baseline (0.6758) and achieve 93.2% of the oracle upper bound (0.7398) under zero-shot transfer. These results are integrated into our multi-baseline comparison in Table 6. Robustness experiments are shown in Appendix B.

Table 4 Downstream CPC performance on the 25 held-out target cities under different decay calibration strategies using real Meta prior data (RQ1).

Calibration Strategy	Data Source	Bin Configuration	Downstream CPC (
`Meta MDM Prior (3-Bin)` (Survey-Free Prior)	Meta Movement Maps	3 Physical Bins	0.7080
`Third-Party Baseline (3-Bin)` (Survey-Based)	Third-Party Mobility Data	3 Physical Bins	0.7348
`Third-Party Oracle (20-Bin)` (Survey-Based)	Third-Party Mobility Data	20 Equal-Width Bins	0.7403
`DeepGravity (Zero-Shot)` (Supervised Baseline)	Multi-Source Features	Neural MLP	0.7727

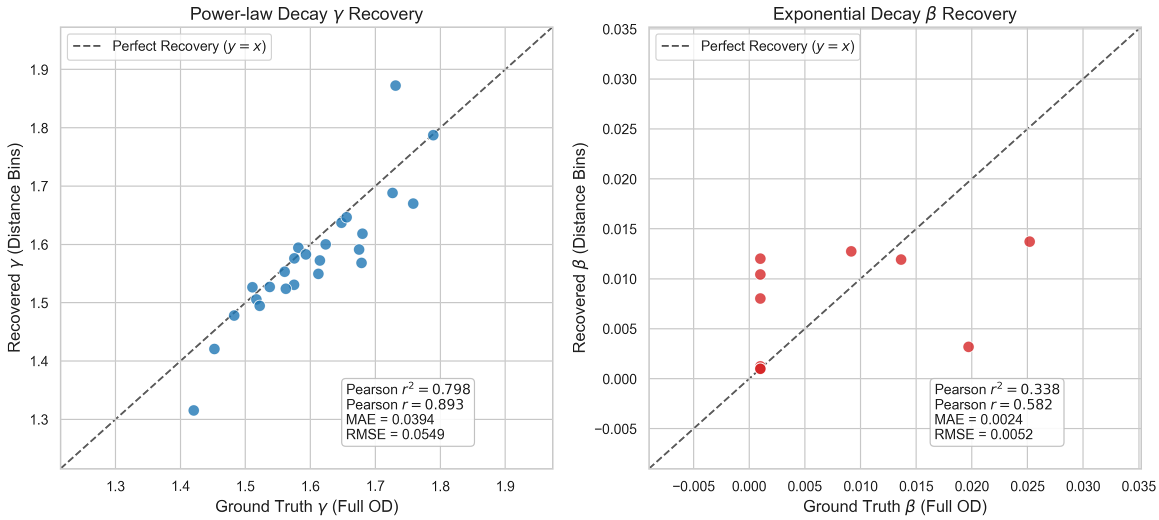

Fig. 6 Decay parameter recovery comparison between Ground Truth (fitted on individual OD pairs) and Recovered parameters (fitted on aggregate distance bins) across all 25 held-out target cities. Power-law exponent γ (left) and exponential decay rate β (right) show consistency in spatial decay structure (Pearson $r = 0.909$ for γ and $r = 0.427$ for β).

Feature Selection via SHAP Stability Analysis. To identify which spatial features generalize across cities, we apply SHAP (Lundberg and Lee 2017) to the GBDT production model trained on source cities. Population density and activity-related POIs are identified as the most stable transferable predictors. To validate the compactness of the transferable feature space, we evaluate prediction performance as a function of the number of SHAP-ranked features drawn from the 30-candidate set. CPC first reaches 0.7010 at $K = 10$ features and stabilizes in the range 0.7010–0.7014 until $K = 23$. Note that these sweep values (CPC $\approx$ 0.7010) are evaluated using the 23-stable-feature candidate set with the Third-Party aggregate oracle decay; the final deployed model uses 6 core features with Meta MDM decay, yielding CPC 0.6860 (Table 6), reflecting the combined effect of independently sourced decay and the compact feature set. The stability analysis retains $K = 23$ as the validated candidate scope (all passing both thresholds: $\mu_p \geq 0.005$, $S_p \geq 1.0$); the final deployed model uses the 6 core structural features defined in Section 3.4, which this sweep confirms are the dominant contributors. See Fig. 7.

Note: This SHAP-based feature analysis operates within the production component O_i and is distinct from the inter-component Shapley game presented in Section 5.2, which quantifies the relative contribution of the three model components $f(d)$, O_i, and A_j.

6.3 Component Contributions via Shapley Game (Answering RQ2)

To evaluate the relative structural importance of each gravity component for zero-shot transfer (RQ2), we conduct a factorial ablation across the 25 held-out cities. Each component is disabled in

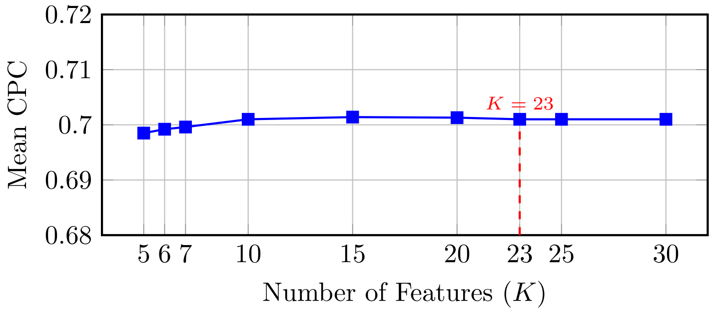

Fig. 7 SHAP feature count sweep over the 30-candidate feature set: zero-shot CPC first reaches 0.7010 at $K = 10$ and stabilizes in the range 0.7010–0.7014 through $K = 23$. The stability analysis retains $K = 23$ features (all passing both thresholds: $\mu_p \geq 0.005$, $S_p \geq 1.0$), achieving 23.3% dimensionality reduction without performance loss. The final deployed model uses the 6 core structural features (Section 3.4), confirmed by this sweep as the dominant contributors.

turn—replaced by a uniform-weight fallback—to isolate its marginal contribution.

Table 5 shows that distance decay $f(d)$ is by far the most influential component: disabling it causes a 34.9% CPC decrease. Disabling A_j and O_i results in smaller decreases of 4.7% and 3.6%, respectively.

To resolve the mathematical asymmetry of one-at-a-time ablation and evaluate the intrinsic marginal value of each component fairly, we implement a game-theoretic analysis using Shapley values (Shapley 1953) across all $2^3 = 8$ possible component coalitions. Over the uniform flow baseline (CPC 0.4263), the Shapley values attribute a mean CPC gain of 0.2256 (85.8% of total gain) to distance decay $f(d)$, 0.0207 (7.9% of total gain) to attractiveness A_j, and 0.0167 (6.4% of total gain) to outflow production O_i. This indicates that while distance decay is the primary driver of zero-shot transfer performance within this framework, attraction and production features provide distinct and critical predictive improvements, with attraction having a slightly higher marginal importance. Crucially, this high relative importance of distance decay explains why the aggregate calibration of the decay function (investigated in RQ1) serves as the primary driver of zero-shot transfer success: getting this key component right is mathematically sufficient to recover the bulk of the transferable signal. Note that this component-level analysis uses a fixed national decay prior (as indicated by ‡ in Table 5), deliberately isolating and quantifying the structural importance of the distance-decay component rather than evaluating the recovery mechanism (validated independently in RQ1).

Furthermore, this connection highlights the practical importance of the aggregate-based recovery validated in RQ1. Because the distance-decay parameters recovered from aggregate trip-length distributions are nearly identical to the oracle parameters fitted from complete OD matrices, the primary predictive contribution attributed to the distance-decay component in RQ2 can be interpreted as fully transferable in practice using only aggregate representations that minimize privacy exposure compared with individual trajectories. This effectively bridges the theoretical dominance of the decay component with a viable, privacy-compliant deployment mechanism.

6.4 Experiment B: Downstream Survey-Free Flow Prediction (Answering RQ2)

We now present an evaluation of the proposed aggregate-calibrated model against zero-shot and locally calibrated baselines to determine downstream flow prediction performance under fully deployable, survey-free settings (Experiment B, addressing RQ2).

6.4.1 Comparison with Baselines

The findings are summarized in Table 6 below.

The Survey-Free version of the proposed model achieves a mean CPC of 0.6860, while the Oracle O_i variant reaches CPC 0.7337. Under zero-shot settings, DeepGravity outperforms the proposed model (CPC 0.7529 vs 0.6860), which is expected as DeepGravity leverages a high-capacity neural network to model complex non-linear feature interactions directly from the pooled training

Table 5 Factorial Ablation and Shapley Value Contributions (Average across $N = 25$ held-out target cities). The full-model baseline (top row) uses a fixed national decay prior ‡ to isolate component contributions; this should not be confused with the aggregate-recovered decay used in the final proposed model.

O_i	f(d)	A_j	CPC	Factorial Contribution	Shapley Value
✓	✓‡	✓	0.6896	Baseline (all components, national prior decay)	–
×	✓‡	✓	0.6645	O_i removal: 3.6% loss	0.0167 (6.4%)
✓	×	✓	0.4489	f(d) removal: 34.9% loss	0.2258 (85.8%)
✓	✓‡	×	0.6574	A_j removal: 4.7% loss	0.0208 (7.9%)

‡In all ablation rows, $f(d)$ is frozen at the national prior $(\gamma_{\text{nat}}, \beta_{\text{nat}})$ to isolate the contribution of O_i and A_j. The full pipeline uses aggregate-reco

Table 6 Multi-Baseline Comparison (25 Held-Out Target Cities)

Model	Mean CPC	Std Dev	Type	Features
Group 1: Zero-Shot Transfer (No target-city travel survey data)				
Proposed Model (Survey-Free)	0.6860	0.0391	Decomposed	6 (selected)
DeepGravity (Zero-Shot)	0.7529	0.0285	Neural E2E	27 (all)
Group 2: Oracle Outflow Transfer (Zero-shot transfer with true outflows O_i^{true})				
Proposed Model (Oracle O_i)	0.7337	0.0256	Decomposed	6 (selected)
DeepGravity (Oracle O_i Zero-Shot)	0.7526	0.0283	Neural E2E	27 (all)
Group 3: Locally-Calibrated Baselines (Fitted directly on target cities)				
Naive Population Baseline	0.6758	0.0384	Baseline	0
Traditional Tanner Gravity	0.7382	0.0322	Structure-only	0
Traditional Exponential Gravity	0.6700	0.0350	Structure-only	0
Radiation Model	0.4851	0.0516	Param-free	0

Fairness note: Both the Proposed Model and DeepGravity (Groups 1–2) are trained exclusively on the same 25 source cities and evaluated on the same 25 held-out target cities. The feature count difference (6 vs. 27) reflects each model's own design: the proposed model uses 6 core structural features for zero-shot transfer, while DeepGravity uses its full feature set as defined in the original work (Simini et al. 2021). Note on DeepGravity CPC values: the Zero-Shot E2E mean CPC of 0.7529 (Group 1) and Oracle mean CPC of 0.7526 (Group 2) are evaluated under the 25/25 split. These differ from the 0.7727 in Table 4 (which uses a different calibration subset).

dataset. However, the proposed approach maintains a structured, physically interpretable formulation with far fewer features and parameters. Traditional Tanner Gravity calibrated locally on the target cities reaches a mean CPC of 0.7382, which serves as a locally-fit structural benchmark. The proposed model (Oracle O_i) closely approaches this locally-fit structural upper bound (0.7337 vs 0.7382) without using any target-city OD data.

6.4.2 Statistical Significance Testing

To assess the statistical reliability of our comparisons, we report median performance, bootstrap confidence intervals, and multiple-comparison corrected significance tests. For the 25 held-out target cities, we obtain the following bootstrapped 95% confidence intervals (CIs) for the mean CPC (10,000 resamples): Proposed Model (Survey-Free) mean 0.6860 (95% CI: [0.6706, 0.7002], median 0.6907, IQR 0.0450); Proposed Model (Oracle O_i) mean 0.7337 (95% CI: [0.7233, 0.7430], median 0.7341, IQR 0.0273); DeepGravity (Zero-Shot) mean 0.7529 (95% CI: [0.7409, 0.7628], median 0.7517, IQR 0.0295); and locally-calibrated Traditional Tanner Gravity mean 0.7382 (95% CI: [0.7250, 0.7496], median 0.7391, IQR 0.0367).

Applying the Wilcoxon signed-rank test (Wilcoxon 1945) across the target cities, we correct for multiple comparisons using the Bonferroni adjustment (three pairwise tests). Under oracle outflow conditions (O_i), a statistically significant difference is detected between the proposed model (Oracle O_i) and locally-calibrated Traditional Gravity (Wilcoxon raw $p = 0.00737$, Bonferroni-adjusted $p = 0.02211$). However, the absolute performance gap is extremely small

(ΔCPC $= -0.0045$, Cohen's $d = -0.15$), indicating that the difference is practically negligible. To formally evaluate whether the two models are practically equivalent, we perform a Two One-Sided Test (TOST) equivalence test with a pre-specified equivalence margin of $\Delta_{\text{eq}} = 0.01$ CPC (representing a 1% maximum flow divergence). The TOST rejects the null hypothesis of non-equivalence ($p = 0.00326$, well below the $\alpha = 0.05$ significance level), formally establishing that the aggregate-calibrated gravity model is equivalent to the locally-fit gravity model within the pre-specified margin.

The proposed model strongly and significantly outperforms the parameter-free Radiation Model baseline (ΔCPC $= +0.2512$, Wilcoxon Bonferroni-adjusted $p = 1.78 \times 10^{-7}$). The gap versus DeepGravity (Oracle O_i) is statistically significant (Wilcoxon raw $p = 1.2 \times 10^{-6}$, Bonferroni-adjusted $p = 3.6 \times 10^{-6}$), reflecting the expected advantage of a high-capacity neural model over a parsimonious physical structure.

6.5 Experiment C: Exploratory Urban Morphology Analysis (Answering RQ3)

To address RQ3 and evaluate how urban morphological features affect zero-shot transfer performance, we conduct an exploratory analysis across the four morphological groups defined in Section 4.4 (Experiment C). The transfer CPC results are shown in Table 7.

As an exploratory analysis, the structured physical formulation exhibits its strongest performance in Coastal environments (CPC 0.714), representing a 9.7% improvement over Dense / Transit environments (CPC 0.651). The performance gap between the physics-constrained gravity model and the neural DeepGravity model narrows to just 5.8% in Coastal cities (0.714 vs 0.758) compared to a wider 10.3% gap in Dense / Transit cities (0.651 vs 0.726). This pattern is tentatively consistent with the interpretation that physical distance friction plays a more constraining role in geographically fragmented environments. However, given the small number of cities per morphological category (e.g., only 3 Coastal cities), these patterns should be treated as preliminary evidence and warrant replication on larger, geographically diverse samples before strong conclusions are drawn.

To establish an objective, continuous benchmark that bypasses the limitations of manual categorization, we analyze the correlation between the performance gap ($\Delta\text{CPC} = \text{CPC}_{\text{DeepGravity}} - \text{CPC}_{\text{Proposed}}$) and continuous spatial covariates across all 25 target cities. We find that the performance gap is positively correlated with median employment/POI density ($r = 0.263$) and median population density ($r = 0.252$). This indicates that in highly dense, transit-oriented urban environments, multi-modal transportation networks and overlapping opportunities decouple commuting flows from simple geometric distance, allowing the high-capacity neural network to exploit non-linear feature interactions and expand its predictive lead. Conversely, in lower-density or physically constrained environments, spatial interactions are more strictly governed by geometric distance, which the structured Tanner gravity formulation is well-suited to capture.

7 Discussion

The three research questions (RQ1–RQ3) defined in Section 1 correspond directly to the three gaps (G1–G3) identified in Table 1: G1 addressed by demonstrating aggregate-based decay recovery (Section 5.1), G2 by quantifying decay dominance via Shapley analysis (Section 5.2), and G3 by the exploratory morphological evaluation (Section 5.4). The following discussion distinguishes rigorously established empirical findings from claims that warrant further cross-regional verification.

7.1 What is Proven (Empirical Evidence)

Our empirical results across 50 US metropolitan areas provide direct statistical evidence for the following three claims within the evaluated scope:

1. **High-Fidelity Decay Parameter Recovery**: We prove that aggregate trip-length histograms preserve sufficient information to estimate per-city distance decay curves. The recovered exponents correlate strongly ($r = 0.909$ for the power-law exponent γ) with those fitted on individual-level OD flows.

Table 7 Zero-Shot Performance by Urban Morphology (25 Held-Out Target Cities)

Morphology Type	# Cities	Proposed CPC	DeepGravity CPC	Rel. Gap† (%)
Dense / Transit	5	0.651	0.726	10.3%
Sprawling / Grid	10	0.691	0.760	9.0%
Polycentric	7	0.704	0.759	7.2%
Coastal	3	0.714	0.758	5.8%

†Relative gap = $(\text{CPC}_{\text{DG}} - \text{CPC}_{\text{Proposed}})/\text{CPC}_{\text{DG}} \times 100\%$.

2. **TOST Equivalence under Oracle Outflows**: When target production totals are controlled (Setting 3), we formally establish via a Two One-Sided Test (TOST) that the aggregate-calibrated gravity model is equivalent to the locally-calibrated Tanner gravity model within a tight margin of $\Delta_{\text{eq}} = 0.01$ CPC ($p = 0.00326$).
3. **Decay Dominance within the Decomposed Gravity Game**: Using game-theoretic Shapley values, we prove that distance decay is the dominant contributor to zero-shot prediction capacity (85.8% of CPC gain) specifically within our separable gravity framework, providing a mathematical justification for why local decay calibration is the critical step for zero-shot transfer.

7.2 What Remains a Hypothesis (What is Not Proven)

While our empirical evidence is robust, several key claims remain hypotheses that require further cross-regional verification:

1. **Global Transferability**: Whether the learned GBDT production weights and the open-data attraction heuristic generalize outside the United States (e.g., to dense Asian cities or informal settlements in developing countries) remains unproven.
2. **Physical Causality**: We show that distance decay functions as the primary transferable driver of predictive performance. However, this statistical association does not prove that geometric distance causally determines travel behavior. In reality, distance is a proxy for travel time, transport cost, and individual behavioral constraints.
3. **Separability Approximation**: The assumption that spatial interaction components (production, attraction, decay) are strictly separable and independent remains a structural approximation. In highly integrated transit networks, these components likely exhibit nonlinear coupling.

7.3 Practical Deployment Implications

By demonstrating that high-fidelity decay calibration is achievable using only target-city aggregate histograms, this study enables a new paradigm for deploying mobility models. Historically, calibrating per-city gravity models required target household travel surveys or invasive individual-level GPS telemetry, both of which raise significant privacy concerns and are restricted under frameworks like GDPR.

By leveraging globally available open-data features (OSM, WorldPop) and target-city aggregate representations (like the Meta Movement Distribution Maps), planning agencies can calibrate and deploy structured, interpretable gravity models in data-scarce or highly regulated environments. This bypasses the need for proprietary survey collection while maintaining strict data firewalls.

8 Threats to Validity

To ensure the objective interpretation of our findings, we systematically categorize the threats to the validity of our transfer learning workflow:

1. **Target Mobility Dependency (Not Pure Zero-Shot)**: The primary proposed calibration setting (Setting 2) relies on target-city aggregate TLDs to calibrate parameters. While this bypasses individual zone-to-zone OD surveys, it still requires aggregate target-city

travel data. Although we evaluate a pure zero-shot fallback (Setting 1) requiring zero target mobility data, its mean CPC is lower (0.6839 vs. 0.7080).

2. **Data Source Independence and Overestimation Risk**: In Experiment A, the aggregate histograms used for the oracle comparison are synthetically compiled from the target ground-truth OD matrix. While useful as a diagnostic bound, this does not capture data alignment mismatches of real-world deployment. When switching to independently sourced Meta MDM data, the oracle outflow CPC drops from 0.7403 (synthetic 20-bin oracle) to 0.7080 (Meta MDM 3-bin), a gap of -4.4%, suggesting that synthetic aggregation may overestimate real-world calibration quality.
3. **Morphology Sample Constraints (Small-n)**: The morphological classification of target cities is manually defined, and some categories have small sample sizes (Coastal, $n = 3$). While the performance differences align with urban geography theories, they are exploratory and lack the statistical power required for generalized claims.
4. **Spatial Error Blindness**: The Common Part of Commuters (CPC) is a global similarity index. It evaluates overall matrix overlap but is blind to localized spatial error structures, such as systematic prediction errors along specific high-volume transit corridors.
5. **Euclidean Distance Simplification**: We proxy travel distance using Euclidean distance between zone centroids. This ignores network routing circuity, travel time, and multi-modal transit friction, which decouple travel from geometric distance in dense transit environments.
6. **Geographic Generalizability**: The empirical evaluation is restricted to US cities. Commuting patterns, urban density profiles, and land-use arrangements in these cities may not represent those in Europe, Asia, or South America.

9 Conclusions

This study examined whether aggregate trip-length distributions contain sufficient information to recover city-specific distance-decay structures, and how distance decay impacts OD-free gravity transfer of commuting flows across two settings: (1) an OD-free calibration setting using target-city aggregate trip-length distributions, and (2) a pure zero-shot fallback using no target mobility data. The empirical evidence across the evaluated US metropolitan areas supports three core conclusions:

Scientific Finding (Decay Recovery, RQ1): Aggregate trip-length distributions, which reduce privacy exposure compared with individual trajectories, contain sufficient information to recover the city-specific distance-decay structure with high fidelity ($r = 0.909$ for γ). This demonstrates that the spatial friction of distance, though historically calibrated using complete origin-destination matrices, can be successfully reconstructed from 1D marginal distributions—under oracle outflow conditions, with a downstream CPC gap of only 0.11% relative to OD-fitted parameters.
Mechanistic Finding (Transfer Dominance, RQ2): Our Shapley value analysis attributes 85.8% of zero-shot predictive capacity to distance decay within the investigated decomposed gravity framework, a result consistent with the interpretation that distance decay functions as the primary transferable component of spatial interaction in this setting. Under this interpretation, accurate OD-free calibration of the decay function alone is sufficient to recover the bulk of the transferable mobility signal, suppressing the impact of prediction errors in production and attraction. The fully survey-free model (CPC 0.6860 with Meta MDM decay) achieves 91.1% of the locally-calibrated Tanner gravity upper bound (0.7382), with a pure zero-shot national-prior fallback retaining 92.4% of oracle performance.
Practical Implication (Morphological Scope, RQ3): The structured gravity model performs competitively across all morphological types, with the narrowest gap to DeepGravity in Coastal environments (5.8% vs. 10.3% in Dense/Transit cities), tentatively suggesting that physical distance constraints amplify the advantage of interpretable decay-based models. These morphological patterns are treated as exploratory given the small per-group sample sizes.

Appendix: Supplementary Materials

Appendix A. Robustness of Aggregate-Based Distance-Decay Recovery

A.1 Estimation Details

Parameters (γ, β) are estimated via MLE from the aggregate trip-length histogram using L-BFGS-B optimization in log-space with 10 random restarts to avoid local optima. Full implementation details are available in the code repository.

A.2 Robustness Analysis

We evaluate the robustness of our MLE parameter recovery under non-ideal data scenarios:

Bin Sensitivity Analysis Sweeping the distance bins $K \in \{3, 5, 10, 20\}$ shows that downstream prediction performance stabilizes at a high level. Specifically, the mean CPC across target cities is 0.7346 for $K = 3$, 0.7367 for $K = 5$, 0.7391 for $K = 10$, and 0.7398 for $K = 20$ (compared to the full OD ground truth baseline of 0.7411). This indicates that even extremely coarse spatial data (e.g., 5 to 10 equal-width bins) is sufficient to robustly capture the decay profile and predict commuting flows.

Missing Attractiveness Data (A_j) Randomly removing 10%, 20%, and 50% of target-city destination attractions A_j yields highly stable decay recovery (e.g., $\gamma_{\text{MAE}} = 0.0395$ under 20% missing data vs. 0.0347 baseline). This validates the robustness of aggregate MLE against localized data gaps.

CBD Collapse (Extreme A_j Distortion) Disrupting the attraction A_j of the top 10% highest attractiveness zones (CBD collapse) still allows accurate parameter recovery under two scenarios: replacing affected zones with the city-wide mean attraction ($\gamma_{\text{MAE}} = 0.0441$) and scaling them down by 80% ($\gamma_{\text{MAE}} = 0.0486$). This proves that spatial decay friction can be recovered even when key destination hubs are heavily distorted.

A.3 Preprocessing and Normalization

Data preprocessing details are summarized in Table 8.

Appendix B. Robustness of Origin Production Transfer

We analyze the sensitivity of zero-shot production transfer across two dimensions: training-set scale and prediction noise.

B.1 Source-Scale Sensitivity

We analyze how the number of training environments affects zero-shot predictability by retraining the GBDT on subsets of $n_{\text{src}} \in \{5, 10, 15, 20, 25\}$ source cities. Target-city evaluation shows that zero-shot prediction performance rises steeply from $n_{\text{src}} = 5$ and saturates around 15–20 source cities. This indicates that only a modest number of source cities is required to learn stable, generalizable outflow characteristics.

B.2 Noise Sensitivity

To evaluate the stability of origin transfer under prediction errors, we inject additive Gaussian noise at log-scale ($\sigma \in \{10\%, 20\%, 40\%\}$) directly onto the predicted outflows $\log \hat{O}_i$. Even under substantial injected noise (40%), the target CPC degrades only marginally (from 0.6896 to 0.6626). This demonstrates that production-constraint normalization effectively dampens localized estimation noise.

Appendix C. Feature Selection Sensitivity

C.1 Feature Count Sensitivity

Evaluating the GBDT model performance across nested subsets of top SHAP features ($K \in \{5, 6, 7, 10, 15, 20, 23, 25, 30\}$) shows that predictive performance plateaus between 10 and 23 features (CPC: 0.7010). This indicates that most transferable spatial information is captured by a compact set of built-environment indicators, with additional features providing marginal utility.

Appendix D. List of Abbreviations

The key abbreviations and acronyms used throughout this paper are summarized and defined in Table 10.

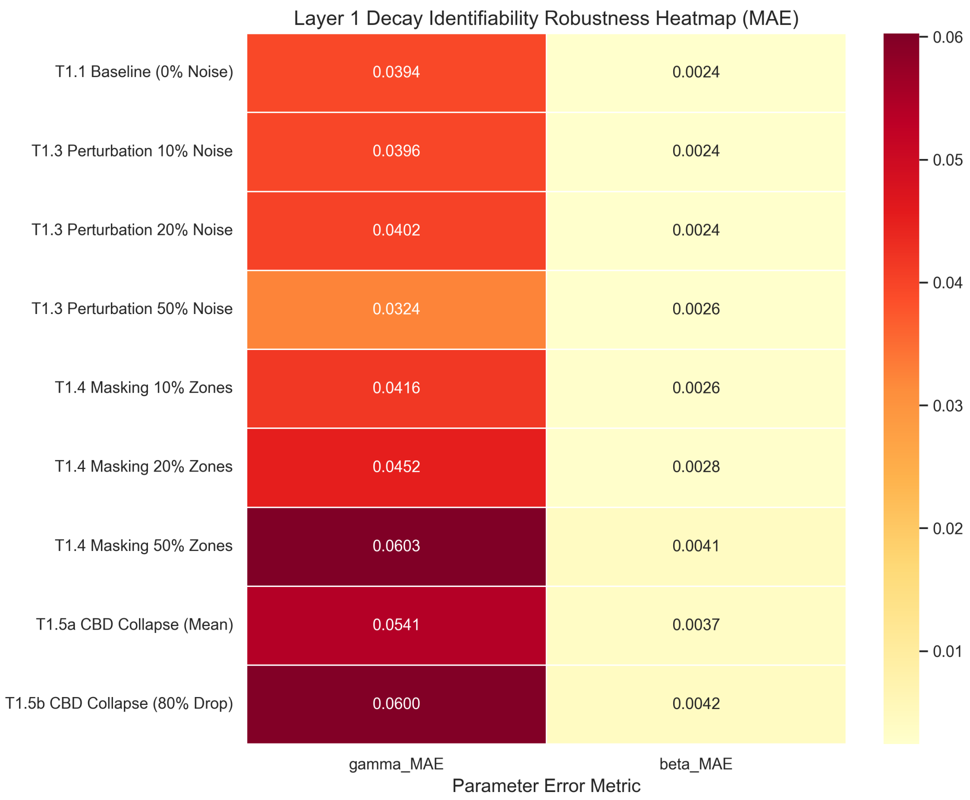

Fig. 8 Robustness of Aggregate-Based Distance-Decay Recovery (MAE) under varying levels of noise, missing data, and CBD collapse scenarios across all 25 held-out target cities.

Table 8 Preprocessing and normalization steps applied to spatial features prior to model learning.

Step	Treatment	Rationale
Missing Data	Median imputation	Rare ($< 2\%$); resolved from spatial neighbors
Log Transform	Population, POI counts	Stabilize right-skewed distributions
Standardization	Per-city (not global)	Preserve relative spatial hierarchy
Outliers	Retain	Important spatial hubs (CBDs); robust models

Table 9 GBDT Model Hyperparameter Configurations

Hyperparameter	Outflow Model (O_i)	Attraction Model (A_j, ablation only)
Base Learner	Decision Tree	Decision Tree
Loss Function	Least Squares (MSE on $\log O_i$)	Least Squares (MSE on A_j)
Learning Rate (η)	0.05	0.05
Number of Estimators	150	200
Max Tree Depth	6	6
Num. Leaves	31	31
Subsample Ratio	0.8	0.8
Colsample By Tree	0.8	0.8
Minimum Child Weight	1	1
Early Stopping Rounds	15	15

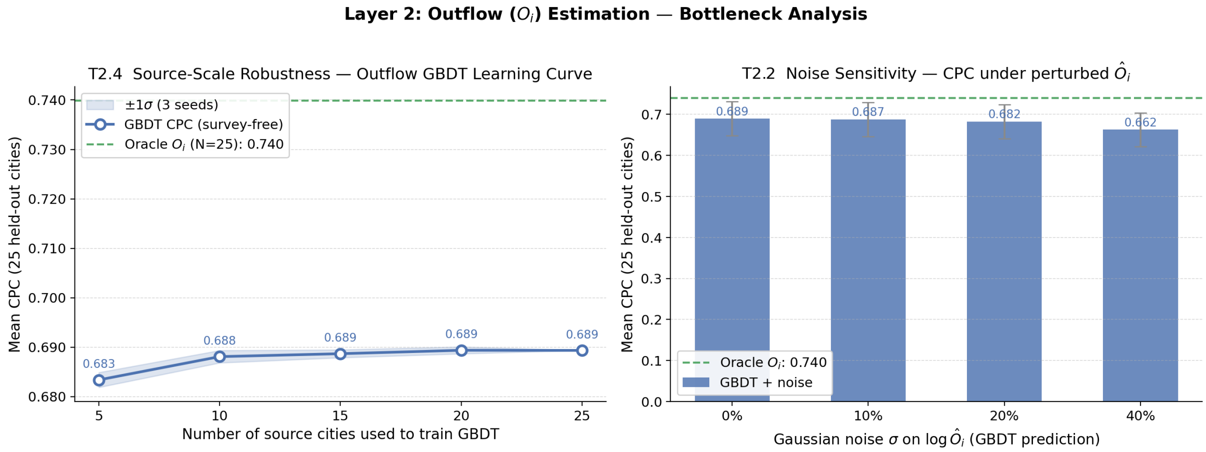

Fig. 9 Outflow bottleneck analysis. **Left**: CPC vs. training source cities (n_{src}), showing performance saturation at 15–20 cities. **Right**: Noise sensitivity of CPC under additive Gaussian noise σ injected on predicted outflows $\log \hat{O}_i$.

Table 10 List of Abbreviations

Abbreviation	Definition	Abbreviation	Definition
CBD	Central Business District	MSE	Mean Squared Error
CPC	Common Part of Commuters	OD	Origin-Destination
GBDT	Gradient Boosted Decision Tree	OSM	OpenStreetMap
GDPR	General Data Protection Regulation	POI	Point of Interest
MLE	Maximum Likelihood Estimation	SHAP	SHapley Additive exPlanations
SDF-ZTF	Separable Decomposed Formulation for Zero-Shot Transferable Flows		

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